at the world down the barrel of a gun! What a horrifically narrow and disgusting view of the world! If you land on a distant planet full of aliens, there is a chance that they will be *friendly* towards you – but apparently this doesn't figure in John's world! It's also unfathomable to me how '3D first person shooter' became a genre. All the 'Halos' and 'Call of Dutys' are essentially the same game – you're supposed to shoot everything that moves! We didn't have this situation in the days of the Spectrum, where, though similarities did exist, they weren't actually the same game. And that was on a machine with 128KB of RAM! So, the upshot is.....the less of his games that are actually played, the better. Marvel at the graphics if you want, feel awed by what the graphics engine is capable of, but don't buy them. I'm also convinced that all these games have their roots in US military training, i.e., that they were designed to train real soldiers for battle, though I have no proof of this. So everyone, please don't play them; tell others not to, and definitely stop giving *money* to these guys—they don't deserve it.

Finally, let's talk about one of the most important developments in gaming today–the Oculus Rift. This is a virtual reality headset, which started as a Kickstarter project and has aroused the interest of Mr Carmack himself, who has promised to develop games for it. I guess that after the Quake 3D engine, which let people get a full 360 degree view of the world, progressing on to something like this was just a natural extension. Games for this are just being developed, and any budding Linux developers can find the information on how to program it, at *oculusvr.com/dk2/*. Your imagination is your only limit since the applications defy any limits—education (imagine studying geography by wandering around distant countries, for example), training (imagine flying a plane by actually being *inside it),* etc, etc. There is no word on whether the Rift is available in India or not, and if not now, then when, but... well, they simply have to let us get our hands on something this cool – not doing so would be criminal! Also, there's no word on the price.

Creation!

Now, after playing all those games, you will, no doubt, be itching to try your own hand at making a couple, so let's talk a little about that. The language I've been busily spending my time learning, for almost the last two years now, is JavaScript. This language runs inside your Web browser, to provide interactivity to Web pages, which is what you'd be familiar with, but can also be bent to drive on-screen graphics. Now, without meaning to knock down your average executable, but since we're all supposed to be 'shifting to the cloud' anyway, I have a strong premonition that this is the way to go. Incidentally, if you'd searched for 'JavaScript game programming' on the Web in 2010, you would have got precious little, but if you do the same today, you'll get a number of books on Amazon. This, obviously, hints at the fact that there's a future in this.

Let's start by getting into the journey of a pixel, from the depths of your browser program, right onto the screen. I have read, more than once, how HTML 5 will kill Flash, and the arguments for that appear to be quite legitimate. Also, games delivered on the Web have the huge advantage of rewriting the supply chain –- if your game delivers to the browser and plays when someone goes to a URL, instead of shipping on a CD or DVD, you cut out middlemen, publishers, distributors, etc, all of whom are not keen on giving you, the developer, your royalty anyway! Added to that is the fact that now, fibre-to-the-home is available from BSNL even in my city (though at a huge price, which is why no one has it yet, but we can all probably expect that the cost will come down in the future).

Now, a browser game written in JavaScript, the interpreter for which is built right into the browser (Firefox), needs only two things to program something in it—an editor (e.g., *gedit*), and Firefox itself. JavaScript offers something called a *<canvas>*, which is exactly what it sounds like, an area of the screen you can paint to. This reminds me of Logo, from way back–a language meant to teach kids programming, by Seymour Papert. It used the concept of a 'turtle'- a robot that moved over a sheet of paper, with a pen inside it. This could be instructed to 'pen up', 'pen down' and so on, and thus create drawings on the paper as it moved. In JavaScript, the basic unit of drawing is a line. After setting up a canvas, you use commands like 'moveto', 'lineto' and 'stroke' to draw stuff on it. When this happens, Firefox draws to the part of your browser that's showing the Web page, which accesses X Windows. X then, according to the driver inside it, writes to video RAM to make that pixel appear on screen.

Let's talk for a bit about the 'network layer' - for those who don't know, X Windows is a 'client server'. You can run X (here called the X server) on one computer, and have it display the result of a program running on another one, *over the network!* To me, this is absolutely astounding, and, as I only tried it out a few months ago, I can confirm that it was, indeed, exactly that. Just for kicks, I ran Blender, the 3D modelling+animation program on a remote machine, and watched it display on the computer I was sitting on. And it worked! Every time I did something to the viewport, the screen updated in 'chunks'- the viewport appeared to be divided into squares, running from left to right, which 'redrew'. Now, this is why when you want to write native games for X, you'll need to learn about 'toolkits' with strange sounding names, and you'll probably need a good bit of C code before you can get a pixel on screen. While this obviously hasn't deterred many folks from trying (see the many games on Steam, and more), why don't we just make all our lives easy and stick to JavaScript for the time being? Not to mention the obvious advantage that your games instantly play cross-platform, i.e., on anything which has a browser (with a recent version of JavaScript), with no added hassle.

However, there are ways to detect if your user is coming from Internet Explorer (this doesn't stop people from using Firefox on the same platform of course), and block the game, if you want to be ethical, which I would highly recommend.